SHRINKING A COMPLETE DEVISSAGE: STANDARD OPEN NEIGHBOURHOODS OF THE BASE, OF THE LAST SCHEME AND OF THE POINT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The notion of a standard shrinking of a complete devissage. Same chapter flat.tex, definition labelled definition-shrink-complete (statement, proof line 1363), together with the item labels item-shrink-base-complete, item-shrink-on-Y-complete and item-shrink-on-X-complete declared inside the counted statement. Fixes what has to be produced in each of the three statements: a standard shrinking whose base lies in a given standard open neighbourhood, or whose last scheme is a given W times the base, or whose source lies in a given open U.
- Base change of a complete devissage at a point. Same chapter flat.tex, label lemma-base-change-complete-at-x, invoked at proof line 1384. Supplies, in part (1), that the base changed data still form a complete devissage at the image point, so that restricting to a standard open neighbourhood of the base gives a standard shrinking.
- The one step shrinking lemmas. Same chapter flat.tex, label lemma-shrink, invoked at proof lines 1384 and 1389 with the item references item-shrink-on-Y and item-shrink-on-X, and again at proof line 1417. Provides one step of shrinking at the level of the last scheme or at the level of the source, which the proof iterates along the devissage in parts (2) and (3); the item labels are navigation pointers outside the counted span and carry the label string itself as their number because they are item labels, not indexed tags.

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Source reference. Original definition 05HL: sources/stacks-flat/pinned-original-source.tex, source lines 1341–1341. Navigation pointer for \ref{definition-shrink-complete} used inside the counted span (line 1363); the argument that consumes it is retained in the span.

Source reference. Original reference 05HD: sources/stacks-flat/pinned-original-source.tex, source lines 985–985. Navigation pointer for \ref{item-shrink-on-X} used inside the counted span (line 1407); the argument that consumes it is retained in the span.

Source reference. Original reference 05HC: sources/stacks-flat/pinned-original-source.tex, source lines 981–981. Navigation pointer for \ref{item-shrink-on-Y} used inside the counted span (line 1389); the argument that consumes it is retained in the span.

Source reference. Original lemma 05HK: sources/stacks-flat/pinned-original-source.tex, source lines 1313–1313. Navigation pointer for \ref{lemma-base-change-complete-at-x} used inside the counted span (line 1384); the argument that consumes it is retained in the span.

Source reference. Original lemma 05HA: sources/stacks-flat/pinned-original-source.tex, source lines 970–970. Navigation pointer for \ref{lemma-shrink} used inside the counted span (line 1385); the argument that consumes it is retained in the span.

Lemma 0.1. *With assumption and notation as in Definition 05HL we have:*

1. *If $S' \subset S$ is a standard open neighbourhood of s, then setting $X' = X_{S'}$, $Z'_k = Z_{S'}$ and $Y'_k = Y_{S'}$ we obtain a standard shrinking.*
2. *Let $W \subset Y_n$ be a standard open neighbourhood of y. Then there exists a standard shrinking with $Y'_n = W \times_S S'$.*
3. *Let $U \subset X$ be an open neighbourhood of x. Then there exists a standard shrinking with $X' \subset U$.*

Proof. Part (1) is immediate from Lemmas 05HK and 05HA.

Proof of (2). For convenience denote $X = Y_0$. We apply Lemma 05HA (05HC) to find a standard shrinking S', Y'_{n-1}, Z'_n, Y'_n of the one step dévissage of $\mathrm{Coker}(\alpha_{n-1})/Y_{n-1}/S$ at y_{n-1} with $Y'_n = W \times_S S'$. We may repeat this procedure and find a standard shrinking $S'', Y''_{n-2}, Z''_{n-1}, Y''_{n-1}$ of the one step dévissage of $\mathrm{Coker}(\alpha_{n-2})/Y_{n-2}/S$ at y_{n-2} with $Y''_{n-1} = Y'_{n-1} \times_S S''$. We may continue in this manner until we obtain $S^{(n)}, Y_0^{(n)}, Z_1^{(n)}, Y_1^{(n)}$. At this point it is clear that we obtain our desired standard shrinking by taking $S^{(n)}$, $X^{(n)}$, $Z_k^{(n-k)} \times_S S^{(n)}$, and $Y_k^{(n-k)} \times_S S^{(n)}$ with the desired property.

Proof of (3). We use induction on the length of the complete dévissage. First we apply Lemma 05HA (05HD) to find a standard shrinking S', X', Z'_1, Y'_1 of the one step dévissage of $\mathcal{F}/X/S$ at x with $X' \subset U$. If $n = 1$, then we are done. If $n > 1$, then by induction we can find a standard shrinking S'', Y''_1, Z''_k, and Y''_k of the complete dévissage $(Z_k, Y_k, i_k, \pi_k, \mathcal{G}_k, \alpha_k, z_k, y_k)_{k=2,\ldots,n}$ of $\mathrm{Coker}(\alpha_1)/Y_1/S$ at x such that $Y''_1 \subset Y'_1$. Using Lemma 05HA (05HC) we can find $S''' \subset S'$, $X''' \subset X'$, Z'''_1 and $Y'''_1 = Y''_1 \times_S S'''$ which is a standard shrinking. The solution to our problem is to take

$$S''', X''', Z'''_1, Y'''_1, Z''_2 \times_S S''', Y''_2 \times_S S''', \ldots, Z''_n \times_S S''', Y''_n \times_S S'''$$

This ends the proof of the lemma. □